

Develop a C program that uses the system calls `open()`, `read()`, `write()`, and `close()` to copy the contents of one file to another. Explain how control transitions between user space and kernel space during execution.

Code:

```
GNU nano 8.7.1
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main() {
    int source, destination;
    char buffer[100];
    int n;

    // Open source file for reading
    source = open("source.txt", O_RDONLY);

    // Open destination file for writing
    // Create it if it does not exist
    // Truncate it if it already exists
    destination = open("copy.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);

    if (source < 0 || destination < 0) {
        printf("Error opening file\n");
        return 1;
    }

    // Read from source and write to destination
    while ((n = read(source, buffer, sizeof(buffer))) > 0) {
        write(destination, buffer, n);
    }

    // Close both files
    close(source);
    close(destination);

    printf("File copied successfully.\n");

    return 0;
}
```

```
GNU nano 8.7.1 source.txt *
This is my Operating Systems Practical - Week 2.
This file is being copied using open(), read(), write(), and close() system calls.
```

Output:

```

priyanka@DESKTOP-HOIS07A:/mnt/c/Users/DELL/OneDrive/Documents/S OS/Practical/Week 2$ pwd
/mnt/c/Users/DELL/OneDrive/Documents/S OS/Practical/Week 2
priyanka@DESKTOP-HOIS07A:/mnt/c/Users/DELL/OneDrive/Documents/S OS/Practical/Week 2$ nano task2.c

priyanka@DESKTOP-HOIS07A:/mnt/c/Users/DELL/OneDrive/Documents/S OS/Practical/Week 2$ nano source.txt

priyanka@DESKTOP-HOIS07A:/mnt/c/Users/DELL/OneDrive/Documents/S OS/Practical/Week 2$ ls
'Task 2.1.docx'  source.txt  task2.c
priyanka@DESKTOP-HOIS07A:/mnt/c/Users/DELL/OneDrive/Documents/S OS/Practical/Week 2$ gcc task2.c -o task2
priyanka@DESKTOP-HOIS07A:/mnt/c/Users/DELL/OneDrive/Documents/S OS/Practical/Week 2$ ls
'Task 2.1.docx'  source.txt  task2  task2.c
priyanka@DESKTOP-HOIS07A:/mnt/c/Users/DELL/OneDrive/Documents/S OS/Practical/Week 2$ ./task2
File copied successfully.
priyanka@DESKTOP-HOIS07A:/mnt/c/Users/DELL/OneDrive/Documents/S OS/Practical/Week 2$ ls
'Task 2.1.docx'  copy.txt  source.txt  task2  task2.c
priyanka@DESKTOP-HOIS07A:/mnt/c/Users/DELL/OneDrive/Documents/S OS/Practical/Week 2$ cat source.txt
This is my Operating Systems Practical - Week 2.
This file is being copied using open(), read(), write(), and close() system calls.
priyanka@DESKTOP-HOIS07A:/mnt/c/Users/DELL/OneDrive/Documents/S OS/Practical/Week 2$ cat copy.txt
This is my Operating Systems Practical - Week 2.
This file is being copied using open(), read(), write(), and close() system calls.

```

System calls provide an interface between user-space programs and kernel-space services. The `open()`, `read()`, `write()`, and `close()` calls allow the program to request file operations from the kernel.